

Functional Requirements :

A) User Requirements Specification (URS):

○ **User Management:**

- Users can register for new accounts with a username, password, and email address.
- Users can log in to their accounts using their username and password.
- Users can edit their profile information, including username, bio, and profile picture.

○ **Content Creation and Sharing:**

- Users can create and post text messages (tweets) with a maximum character limit.
- Users can attach images and videos to their tweets.
- Users can delete their tweets after posting.

○ **Following and Followers:**

- Users can follow other users to see their tweets in their feed.
- Users can see a list of users they follow and their followers.
- Users can unfollow other users.

○ **Feed and Timeline:**

- Users can see a personalized feed of tweets from the users they follow.
- The feed should display the most recent tweets first.
- Users can see options to retweet, like, and reply to other users' tweets.

○ **Search:**

- Users can search for other users and tweets by username, hashtags, or keywords.
- Search results should be displayed in a relevant order based on user activity and tweet content.

B) System Requirements Specification (SRS):

○ **Security:**

- The system must implement secure user authentication and authorization mechanisms.
- User passwords must be stored securely using hashing algorithms.
- The system must protect user data from unauthorized access and modification.

○ **Performance:**

- The system must be able to handle a high volume of users and tweets efficiently.
- Page loading times and response times for user actions should be minimal.
- The system should be scalable to accommodate future growth.

○ **Scalability:**

- The system architecture should be designed to handle increasing user base and data volume.
- The system should be able to add additional servers and resources as needed.

○ **Database:**

- The system must use a database to store user information, tweets, and other relevant data.
- The database must be chosen based on its ability to handle large datasets and high read/write operations.

C) Requirements' Priorities (MoSCoW schema):

○ **Must - Have:**

These are essential requirements that the system must have to function properly. They are non-negotiable and critical for user satisfaction. (e.g., User registration, login, tweet creation, feed display)

○ **Should - Have:**

These are important requirements that enhance the user experience and functionality of the system. They are desirable but not essential for the core functionality. (e.g., Image/video attachments, editing tweets, following/followers)

○ **Could - Have:**

These are desirable features that would add value to the system but are not essential. They can be implemented later based on available resources and priorities. (e.g., Direct messaging, lists, trending topics)

○ **Won't - Have:**

These are features that are considered low priority or outside the scope of the current project. They can be considered for future versions if needed. (e.g., Advanced search filters, complex analytics tools)

Non Functional Requirements :

A) General types / categories of non-functional requirements:

- 1. Performance:** This category focuses on the responsiveness and speed of the system.
- 2. Scalability:** This category addresses the system's ability to handle increasing user base and data volume.
- 3. Availability:** This category ensures the system is accessible and operational for users most of the time.
- 4. Security:** This category focuses on protecting user data, system resources, and overall platform integrity.
- 5. Usability:** This category emphasizes the user interface's ease of use and learnability for a broad audience.
- 6. Maintainability:** This category ensures the system can be easily modified, fixed, and updated as needed.
- 7. Reliability:** This category focuses on the system's ability to consistently perform its functions without failures.
- 8. Recoverability:** This category addresses the system's ability to recover from outages or data loss.

B) Non-Functional Requirements:

- 1. Performance:**
 - 1) The system should display a user's feed within 2 seconds of page load.
 - 2) Users should be able to post a tweet within 5 seconds of composing it.
- 2. Scalability:**
 - 1) The system should be able to handle a daily growth of 10% in new users.
 - 2) The system should be able to scale horizontally by adding new servers.
- 3. Availability:** The system should be available 99.9% of the time (uptime).
- 4. Security:**
 - 1) All user data (passwords, private messages) must be encrypted at rest and transit.
 - 2) User authentication should be multi-factor for logins from unrecognized devices.

- 5. Usability:** 1) The user interface should be intuitive and easy to navigate for all user groups.
- 2) Common actions (tweet creation, reply, search) should be accessible in 2 clicks.
- 6. Maintainability:** The code should be well-documented, modular, and follow coding standards.

C) Fit Criteria for Non-Functional Requirements:

- 1. Performance:** 1) 95% of page loads should render the feed within 2 seconds.
- 2) 90% of tweet posts should be completed within 5 seconds.
- 2. Scalability:** 1) The system should maintain acceptable performance metrics after a 10% user base increase over a week.
- 2) The system should seamlessly integrate additional servers and maintain functionality after deployment.
- 3. Availability:** System uptime should be monitored and documented, with downtime not exceeding 0.1% over a month.
- 4. Security:** 1) Encryption methods used should be industry standards (e.g., AES-256).
- 2) Multi-factor authentication should be mandatory for logins from untrusted devices.
- 5. Usability:** 1) Usability testing should be conducted with representative user groups.
- 2) User interface design should be evaluated to ensure core functionalities are within 2 clicks.
- 6. Maintainability:** Code reviews should be conducted to ensure code quality and adherence to standards.